Transfer-Dependent Frontier Growth: Structural Breaks, Inflation Dynamics, and the COVID-19 Shock in Arunachal Pradesh

S.M. Muzammil Afroz
Subject: Financial Programming
Instructor: Prof. M. Parameswaran
Centre For Development Studies

March 2026

Abstract

This paper studies the growth and inflation experience of Arunachal Pradesh relative to all India and a set of regional and hill-state comparators. The empirical design follows the assignment requirement: estimate trend growth in constant-price GDP and GSDP, test for structural breaks using the Bai-Perron multiple-break framework, compute kinked exponential growth rates, and construct quarterly and annual inflation measures from CPI and implicit deflators. The analysis adds three extensions: a state-comparator module, a sectoral transformation check, and a COVID-19 shock comparison. The central finding is that Arunachal Pradesh is not a simple post-1991 liberalisation acceleration case. Its constant-price GSDP breaks are dated at 1995-96 and 2013-14, not at 1987-88 statehood or the 1991 reforms. Under the Boyce continuity-restricted baseline, trend growth declines from 7.63 percent in 1980-95 to 6.61 percent in 1996-2013 and 5.54 percent in 2014-24. A discontinuous segment-wise OLS sensitivity check preserves the same broad deceleration pattern, but it also implies a large negative fitted level shift at the 1995-96 break, which matters for interpretation. Subperiod evidence is sharper: the pre-liberalisation CAGR is 9.34 percent, above the 5.18 percent recorded during 1991-2003. Inflation evidence shows a frontier-market pattern: Arunachal Pradesh's rural CPI inflation is slightly higher and substantially more volatile than all-India combined CPI inflation, with a particularly large fuel premium. The COVID-19 output shock was negative but not uniquely severe among comparable states; the subsequent recovery was weaker than Assam, Sikkim, Tripura, and Meghalaya. The paper interprets Arunachal Pradesh as a transfer-dependent frontier economy whose growth has been real but only weakly associated with deep industrial transformation.

JEL Classification: O47, R11, E31, C22, O11, H77, O14, E32

Contents

1 Introduction **3**

2 Literature Review **3**

3 Data **4**

4 Methodology **6**

5 Growth Results **9**

6 Structural Transformation **13**

7 Inflation Results **14**

8 COVID-19 Shock and Recovery **19**

9 Robustness and Comparator Evidence **20**

10 Discussion **21**

11 Conclusion **22**

A Additional Tables **24**

B Additional Figures **27**

C Data Limitations **31**

1 Introduction

Economic growth in India is usually narrated around two national questions: when did the long-run acceleration begin, and how much of it can be attributed to the 1991 reforms? That aggregate question is important, but it is incomplete for frontier and hill states. In states such as Arunachal Pradesh, the relevant mechanisms are not only trade liberalisation, industrial licensing, and private investment. They also include state formation, central transfers, infrastructure constraints, thin local markets, and the high transport cost of serving a sparsely populated border economy.

This paper asks whether Arunachal Pradesh's growth and inflation experience resembles the all-India growth-transition story or a more specific frontier-state path. The empirical answer is clear. Arunachal Pradesh grew rapidly, but its growth path does not look like a standard liberalisation acceleration. The Bai-Perron procedure identifies breaks in 1995-96 and 2013-14, whereas the most obvious administrative candidate, the transition to statehood in 1987-88, is not selected. Nor is 1991-92 selected as a growth break for the state. The kinked growth estimates show steady deceleration rather than acceleration: 7.63 percent in 1980-95, 6.61 percent in 1996-2013, and 5.54 percent in 2014-24. Cross-state comparisons reinforce the result. Arunachal Pradesh's pre-1991 CAGR is 9.34 percent, while its 1991-2003 CAGR is only 5.18 percent.

The second part of the paper studies inflation. Arunachal Pradesh's CPI data are rural-only, so the state series is not directly symmetric with all-India combined CPI. Even with this limitation, the pattern is informative. Quarterly rural CPI inflation in Arunachal Pradesh averages 5.90 percent, compared with 5.58 percent for all India, but its standard deviation is much higher. The annual evidence shows large swings: headline inflation is 12.12 percent in 2012-13, 8.91 percent in 2018-19, falls sharply in 2019-20, and then normalises after the pandemic. A cross-state CPI comparison shows that Arunachal Pradesh's mean headline inflation over the available period is close to Tripura and Assam, while its fuel inflation is persistently elevated.

The contribution is therefore both descriptive and interpretive. Descriptively, the paper produces a complete analysis of all-India GDP, Arunachal Pradesh GSDP, CPI inflation, deflator inflation, headline inflation, and core inflation. Interpretively, it argues that Arunachal Pradesh is best read as a transfer-dependent frontier economy rather than a state that followed the all-India liberalisation cycle. The evidence does not deny growth; it changes the explanation of growth. Output expanded, but the timing of breaks, sectoral composition, and inflation volatility point to a state whose macroeconomic performance is shaped by public expenditure, terrain, logistics, and a narrow productive base.

2 Literature Review

The analysis is grounded in four strands of literature. The first studies growth as a sequence of regimes rather than a single average. Hausmann et al. (2004) define growth accelerations as sustained departures from prior trends, while Bai and Perron (1998, 2003) provide the econometric framework for estimating multiple unknown breaks in linear models. Zeileis et al.

(2002) document the *strucchange* implementation used in applied work. Once breaks are identified, two post-break growth specifications matter. The kinked exponential model of Boyce (1986) imposes continuity and interprets breaks as changes in growth rates. A segment-by-segment log-linear trend, by contrast, allows intercept and slope to vary by regime and therefore permits discontinuous level shifts. This distinction is important because Balakrishnan and Parameswaran (2007) discuss Indian growth under the broader structural-change framework in which pure and partial breaks need to be separated carefully.

The second strand concerns India's growth transition. Rodrik and Subramanian (2005) argue that India's growth shift began before 1991, whereas Agarwal and Whalley (2013) caution against a mechanical attribution of growth acceleration to the reforms alone. Balakrishnan and Parameswaran (2007) similarly emphasise the importance of careful growth-rate estimation in the Indian context. This debate matters for Arunachal Pradesh because a post-1991 acceleration should not be assumed; it must be tested against the state series.

The third strand studies regional growth and convergence among Indian states. Sachs et al. (2002) show that regional growth depends on structural conditions, while Bhattacharya and Sakthivel (2004), **?**, and Sodsriwiboon and Kalra (2010) document divergence, polarisation, club formation, and weak spillovers among Indian states. The recent public-finance evidence in Govil and Chauhan (2024) is especially relevant because Arunachal Pradesh's growth cannot be separated from fiscal capacity and central transfers. This literature motivates the comparator set used here: Assam as a Northeast regional baseline, Sikkim as a small Himalayan benchmark, Himachal Pradesh as a hill-state counterfactual outside the Northeast, and Tripura and Meghalaya as additional Northeast comparators.

The fourth strand studies spatial inflation and the COVID-19 shock. Kundu et al. (2018) and Jha and Dhal (2019) show that regional inflation dispersion persists even under a common national monetary policy, especially through food and supply-side channels. **?** and Narayanan and Saha (2021) document the food-supply disruption caused by the pandemic, while **?** show that the COVID-19 economic shock varied across Indian states. These studies are directly relevant for Arunachal Pradesh because terrain, distance from major supply centres, and the thinness of local markets can amplify food and fuel price volatility.

3 Data

The output data come from the EPW Research Foundation India Time Series. All-India GDP is observed at current and constant prices from 1950-51 to 2025-26. Arunachal Pradesh GSDP is observed at current and constant prices from 1980-81 to 2024-25. The assignment requirement is fully covered; the latest available observations are retained where available. Arunachal Pradesh data prior to statehood should be read as the economic series of the territory that became the state, not as evidence of an independent state government's policy regime.

Monthly CPI data are available from January 2011 to December 2025. All-India CPI is available for rural, urban, and combined sectors with major subgroups. Arunachal Pradesh CPI is available only for the rural sector. Its urban CPI is missing for all major groups, and the

sporadic combined values are identical to rural values where reported. Therefore all state-level CPI calculations use Arunachal Pradesh rural CPI. This is not a discretionary choice; it follows from the source data.

Core inflation is computed with the inclusion method. For all India, the core index is constructed from clothing and footwear, housing, and miscellaneous components using the available rural and urban weights. For Arunachal Pradesh, housing is not available in the rural CPI basket, so core inflation uses clothing and footwear and miscellaneous only. The state core measure is therefore narrower and should be interpreted as an indicative non-food, non-fuel inflation measure rather than a perfectly comparable all-India core index. The distortion is not trivial: housing carries zero weight in rural CPI but 21.67 points in the all-India urban core basket used here.

The state-comparator module uses long-run all-state GSDP, NSDP, and CPI data. Table 1 summarises the main series. The chosen comparators are not arbitrary. Assam is the natural regional benchmark; Sikkim captures the small-Himalayan high-growth case; Himachal Pradesh provides a hill-state comparison outside the Northeast; Tripura and Meghalaya add robustness within the Northeast.

Table 1: Main data sources and coverage

Series	Source and unit	Period used	Main limitation
All-India GDP	GDP at current and constant prices, Rs crore	1950-51 to 2025-26	Latest years are provi-sional/revised where applicable
Arunachal Pradesh GSDP	GSDP at current and constant prices, Rs lakh converted as needed	1980-81 to 2024-25	Pre-1987 observations refer to the territory before statehood
All-India CPI	CPI 2012=100, monthly, major groups	Jan. 2011 to Dec. 2025	Core construction uses available CPI groups and weights
Arunachal Pradesh CPI	CPI 2012=100, monthly, rural major groups	Jan. 2011 to Dec. 2025	Urban CPI is absent; combined CPI is not usable
Comparator states	All-state GSDP/NSDP backseries and state CPI	varies by state	Sikkim has no pre-1991 GSDP sample

Notes: The annual inflation table reports the required 2011-12 to 2023-24 window and retains 2024-25 as the latest available extension. Arunachal Pradesh CPI in 2020-21 is affected by missing April and May 2020 observations.

4 Methodology

The baseline growth equation is

$$\ln Y_t = \alpha + \beta t + u_t, \tag{1}$$

where Y_t is real GDP or real GSDP and β is the instantaneous trend growth rate. The compound annual growth rate is computed as

$$\text{CAGR} = \left(e^{\beta} - 1\right) \times 100. \tag{2}$$

Structural breaks are estimated using the Bai-Perron multiple-break framework with up to five breaks and a trimming parameter of 0.10, with model selection by the Bayesian information criterion (Bai and Perron 1998; Bai and Perron 2003; Zeileis et al. 2002).

When breaks are detected, the baseline growth specification is the Boyce kinked exponential model:

$$\ln Y_t = \alpha + \beta_1 t + \sum_{j=1}^{m} \delta_j (t - T_j) D_{jt} + u_t, \tag{3}$$

where T_j is the jth break date and $D_{jt} = 1$ if $t > T_j$. The post-break regime-j growth coefficient is therefore $\beta_1 + \delta_1 + \cdots + \delta_j$. The specification permits the slope to change at a break while maintaining continuity in the level of log output. This is appropriate for trend-growth estimation because structural breaks in growth need not imply jumps in the level of output (Boyce 1986). Heteroskedasticity-robust standard errors for regime-specific slopes are computed with the delta method from the pooled covariance matrix, and bootstrap confidence intervals are reported for the baseline growth rates.

The paper also reports a discontinuous sensitivity specification:

$$\ln Y_t = \alpha_j + \beta_j t + u_t \qquad \text{for } t \in \text{regime } j. \tag{4}$$

Here, each Bai-Perron regime is estimated separately, so both intercepts and slopes may differ across regimes. This segment-wise OLS specification is useful because it produces direct regime-specific standard errors and can reveal fitted level shifts at the break dates. It should not be called the Boyce kinked model, because the point of the Boyce restriction is precisely to rule out such jumps. Following Balakrishnan and Parameswaran (2007), the paper therefore treats the continuous pooled equation as Baseline A and the discontinuous segment-wise regression as Baseline B. Under fixed break dates and an imposed continuity restriction, Baseline A and Baseline B can deliver similar slope estimates while implying different interpretations of what happened at the break itself.

The COVID-19 shock is studied in two ways. First, a 2020-21 pulse dummy is added to the growth model:

$$\ln Y_t = \alpha + \beta_1 t + \sum_{j=1}^{m} \delta_j (t - T_j) D_{jt} + \gamma\, \mathbf{1}\{t = 2020\text{–}21\} + u_t. \tag{5}$$

Second, a more transparent realised-shock metric compares actual real GSDP growth from 2019-20 to 2020-21 with the pre-COVID trend from 2014-15 to 2019-20:

$$\text{Shock}_s = g^{\text{actual}}_{s,\,2019\text{–}20\to 2020\text{–}21} - g^{\text{trend}}_{s,\,2014\text{–}15\to 2019\text{–}20}. \tag{6}$$

The post-COVID recovery CAGR is computed as

$$\text{Recovery CAGR}_s = \left(\frac{Y_{s,\text{latest}}}{Y_{s,2020\text{–}21}}\right)^{1/n_s} - 1, \tag{7}$$

where n_s is the number of years between 2020-21 and the latest available observation for state s. The realised-shock metric is used for cross-state comparisons because it is easy to interpret and less dependent on the parametric break specification.

For price indices, monthly observations are first mapped into Indian fiscal years, where April–June is Q1, July–September is Q2, October–December is Q3, and January–March is Q4. Let $I_{s,g,m}$ denote the CPI index for state or sector s, item group g, and month m. The fiscal-quarter average is

$$\bar{I}_{s,g,fq} = \frac{1}{N_{s,g,fq}} \sum_{m\in fq} I_{s,g,m}, \tag{8}$$

where $N_{s,g,fq}$ is the number of observed months in that fiscal quarter. Quarterly CPI inflation is then computed year-on-year as

$$\pi^{Q}_{s,g,fq} = \left(\frac{\bar{I}_{s,g,fq}}{\bar{I}_{s,g,fq-4}} - 1\right) \times 100. \tag{9}$$

Fiscal-year CPI averages are constructed as

$$\bar{I}_{s,g,f} = \frac{1}{M_{s,g,f}} \sum_{m\in f} I_{s,g,m}, \tag{10}$$

where $M_{s,g,f}$ is the number of observed months in fiscal year f. Annual headline inflation is

$$\pi^{H}_{s,f} = \left(\frac{\bar{I}_{s,\text{general},f}}{\bar{I}_{s,\text{general},f-1}} - 1\right) \times 100. \tag{11}$$

Food and fuel inflation are computed analogously by replacing the general index with the food or fuel group average.

Core inflation follows the inclusion method described below. For all India, the monthly rural core index is

$$C^{R}_{m} = \frac{7.36\, I^{R}_{\text{cloth},m} + 27.26\, I^{R}_{\text{misc},m}}{34.62}, \tag{12}$$

the monthly urban core index is

$$C^{U}_{m} = \frac{5.57\, I^{U}_{\text{cloth},m} + 21.67\, I^{U}_{\text{housing},m} + 29.53\, I^{U}_{\text{misc},m}}{56.77}, \tag{13}$$

and the combined all-India monthly core index is

$$C_m^{AI} = \frac{0.5462 \times 34.62\, C_m^R + 0.4538 \times 56.77\, C_m^U}{0.5462 \times 34.62 + 0.4538 \times 56.77}. \tag{14}$$

This is a population-weighted rural-urban core index, renormalised within the core basket rather than the full CPI basket. The corresponding annual all-India core inflation rate is

$$\pi_{AI,f}^C = \left(\frac{\bar{C}_f^{AI}}{\bar{C}_{f-1}^{AI}} - 1 \right) \times 100, \qquad \bar{C}_f^{AI} = \frac{1}{12} \sum_{m \in f} C_m^{AI}. \tag{15}$$

For Arunachal Pradesh, urban CPI is unavailable and housing is absent from the rural CPI basket. The monthly state core index is therefore

$$C_m^{AR} = \frac{6.44\, I_{\text{cloth},m}^R + 24.70\, I_{\text{misc},m}^R}{31.14}, \tag{16}$$

and annual state core inflation is

$$\pi_{AR,f}^C = \left(\frac{\bar{C}_f^{AR}}{\bar{C}_{f-1}^{AR}} - 1 \right) \times 100, \qquad \bar{C}_f^{AR} = \frac{1}{M_{AR,f}^C} \sum_{m \in f} C_m^{AR}. \tag{17}$$

In the estimation procedure, India fiscal-year core is retained only when all 12 months are present; Arunachal Pradesh fiscal-year core is retained when the available-month count is sufficient to preserve the chain, which leaves 2020-21 in the sample with an explicit caveat. In the Arunachal Pradesh CPI workbook, duplicate all-missing general-index rows appear in early 2013; these rows are dropped before monthly merges so that annual core inflation is not mechanically distorted by duplicated months.

Deflator-based inflation is computed from the ratio of current-price to constant-price output:

$$P_t^D = 100 \times \frac{Y_t^N}{Y_t^R}, \tag{18}$$

where Y_t^N is current-price GDP or GSDP and Y_t^R is constant-price GDP or GSDP. At the annual frequency, deflator inflation is recorded as the log difference

$$\pi_t^{D,A} = 100 \left(\ln P_t^D - \ln P_{t-1}^D \right). \tag{19}$$

Because quarterly GSDP is not directly observed for Arunachal Pradesh, annual current-price and constant-price GSDP are temporally disaggregated separately with the ‘tempdisagg‘ implementation

$$\texttt{td}(Y_t \sim 1,\ \texttt{to = ``quarterly''},\ \texttt{method = ``denton-cholette''}), \tag{20}$$

that is, Denton-Cholette interpolation without a high-frequency indicator, using a constant preliminary series (Sax and Steiner 2013). In the unifying notation of Sax and Steiner, the

quarterly estimate is

$$\hat{\mathbf{y}} = \mathbf{p} + D_{DC}\left(\mathbf{y}_l - C\mathbf{p}\right), \tag{21}$$

where $\mathbf{y}_l$ is the annual series, C is the annual-summing conversion matrix, $\mathbf{p}$ is the constant preliminary quarterly series, and D_{DC} is the Denton-Cholette distribution matrix. The essential accounting restriction is

$$C\hat{\mathbf{y}} = \mathbf{y}_l, \tag{22}$$

equivalently,

$$\sum_{q \in t} \hat{y}_q = Y_t \quad \text{for each fiscal year } t. \tag{23}$$

Operationally, this produces a smooth quarterly path that exactly preserves annual benchmarks while avoiding the initial transient movement associated with the original Denton procedure (Dagum and Cholette 2006; Sax and Steiner 2013). After interpolation, the quarterly deflator is

$$P_q^D = 100 \times \frac{\hat{Y}_q^N}{\hat{Y}_q^R}, \tag{24}$$

and quarterly deflator inflation is computed year-on-year as

$$\pi_q^{D,Q} = \left(\frac{P_q^D}{P_{q-4}^D} - 1\right) \times 100. \tag{25}$$

5 Growth Results

Table 2 reports the structural break dates. All-India GDP has four breaks under the baseline trimming parameter: 1971-72, 1978-79, 2003-04, and 2018-19. The result is consistent with the view that India's growth transition is not reducible to a single 1991 break. For Arunachal Pradesh, the estimated breaks are 1995-96 and 2013-14. Neither 1987-88 nor 1991-92 is selected. This is one of the paper's central empirical results.

The break dates also need an economic reading. The 1995-96 break is best interpreted as a mid-1990s transition within a public-investment and administrative-catch-up growth model rather than as a single reform event. Official Northeast policy documents for 1995-96 emphasised roads, power projects, and region-building institutions under the North Eastern Council, which is consistent with a frontier economy whose growth depended heavily on public expenditure and connectivity support rather than on private industrial take-off (Ministry of Home Affairs, Government of India 1996; Cashin and Sahay 1995). The 2013-14 break is easier to read as the onset of a slower regime. It lines up with the later comparator evidence, where Assam, Sikkim, and Tripura outperform Arunachal Pradesh after 2013, and with a sectoral pattern in which services remain dominant while industry stays modest (NITI Aayog and NCAER 2025). A plausible mechanism is the mid-decade shift in the transfer regime itself: the Fourteenth Finance Commission moved more resources into untied devolution even as it explicitly noted the North-eastern states' heavy dependence on central transfers for capital expenditure and infrastructure (Fourteenth Finance Commission 2015). For a state such as Arunachal Pradesh,

that kind of shift could preserve revenues while weakening the project-specific growth impulse that earlier plan-style and infrastructure-linked support had provided. In that sense, the two breaks point to regime shifts inside the same frontier-state development model: first from early catch-up to slower transfer-supported growth, and then from that phase to a still slower post-2013 regime.

The extra all-India breaks also require interpretation. The short 1971-72 to 1978-79 regime is best read as a disturbed interval rather than as a fully separate long-run epoch. It is a seven-year low-growth segment located between the pre-1971 trend and the post-1978 recovery, and it plausibly captures the macroeconomic turbulence around the Bangladesh war, the first oil shock, and the weakening of the early Green Revolution impulse. That segment disappears once trimming is raised above 0.10, so the economically sensible reading is that the baseline specification is allowed to isolate a short transitional slowdown that more conservative trimming rules absorb into the broader pre-1979 period. This is a plausible reason why the present paper finds four breaks while more aggregated accounts of India's growth transition often emphasise only the later 1978-79 and 2003-04 turning points.

The 2018-19 break is easier to interpret as the consolidation of the late-2010s national slowdown. Its timing is more consistent with the cumulative effects of demonetisation, GST transition costs, NBFC stress, and weakening investment demand than with a single one-year event. In this annual sample, those frictions appear to cumulate into a distinct lower-growth regime only by 2018-19, which is why the break is dated later than in some shorter-window studies that place the slowdown closer to the middle of the decade.

Table 2: Bai-Perron structural break results

Series	Breaks	Break 1	Break 2	Additional breaks
All-India GDP	4	1971 [1969, 1972]	1978 [1976, 1979]	2003 [2002, 2004]; 2018 [2016, 2019]
Arunachal Pradesh GSDP	2	1995 [1994, 1996]	2013 [2012, 2014]	none

The baseline kinked growth estimates in Table 3 translate the break dates into regime growth rates. All-India GDP growth rises from 3.88 percent in 1950-71 to 5.53 percent in 1979-2003 and 6.92 percent in 2004-18, before moderating to 4.93 percent in the post-2018 regime. Arunachal Pradesh shows the opposite shape. Its trend growth rate is highest in the first regime, at 7.63 percent, then declines to 6.61 percent and 5.54 percent. The state therefore grows rapidly, but the evidence is of gradual deceleration rather than post-reform acceleration.

Table 3: Kinked exponential trend growth rates

Series	Period	β	CAGR	Bootstrap 95% CI	Years
All-India	1950-1971	0.03802	3.88	[3.74, 4.00]	22
All-India	1972-1978	0.02476	2.51	[2.02, 3.01]	7
All-India	1979-2003	0.05378	5.53	[5.36, 5.68]	25
All-India	2004-2018	0.06694	6.92	[6.65, 7.17]	15
All-India	2019-2025	0.04808	4.93	[2.91, 5.55]	7
Arunachal Pradesh	1980-1995	0.07352	7.63	[6.98, 8.15]	16
Arunachal Pradesh	1996-2013	0.06403	6.61	[6.19, 7.06]	18
Arunachal Pradesh	2014-2024	0.05395	5.54	[5.04, 6.33]	11

Table 4 shows what changes when continuity is relaxed. For Arunachal Pradesh, the segment-wise OLS sensitivity check yields CAGR estimates of 8.65, 7.19, and 4.56 percent across the three regimes. That is not a reversal of the baseline story: growth is still strongest in the earliest regime and weakest after 2013. The important difference is inferential rather than directional. Once a discontinuous specification is allowed, the 1995-96 break is associated with a fitted level shift of about -17.75 percent, while the 2013-14 break is associated with a smaller positive fitted shift of 3.49 percent. In other words, Baseline B tells a mixed story of slower later growth plus possible jumps in the fitted level of output at regime boundaries. Baseline A is therefore retained as the main specification for trend-growth interpretation, while Baseline B is read as evidence that the 1995 break may bundle a level re-alignment with a slope change.

Table 4: Baseline A and Baseline B growth-rate comparison

Series	Period	A CAGR	B CAGR	B robust SE	B level shift
All-India	1950-1971	3.88	3.92	0.00058	n.a.
All-India	1972-1978	2.51	4.74	0.00326	-6.39
All-India	1979-2003	5.53	5.56	0.00041	-8.68
All-India	2004-2018	6.92	6.80	0.00068	4.59
All-India	2019-2025	4.93	6.57	0.01006	-8.42
Arunachal Pradesh	1980-1995	7.63	8.65	0.00171	n.a.
Arunachal Pradesh	1996-2013	6.61	7.19	0.00187	-17.75
Arunachal Pradesh	2014-2024	5.54	4.56	0.00280	3.49

Notes: CAGR entries are in percent. Baseline A is the continuity-restricted Boyce pooled regression. Baseline B is segment-wise OLS estimated separately within each Bai-Perron regime. "B level shift" reports the fitted percentage jump at the start of the regime relative to the previous regime's projected level.

The subperiod comparison in Table 5 makes the interpretation more concrete. Arunachal Pradesh grows at 9.34 percent during 1980-91, much faster than Assam, Himachal Pradesh, Tripura, and Meghalaya. During 1991-2003, however, its growth falls to 5.18 percent. The state does grow faster again during 2003-13, but the post-2013 regime slows to 5.58 percent, below Assam, Sikkim, and Tripura. This trajectory does not fit a simple liberalisation story. It is closer to a frontier-state path where early public investment, administrative consolidation, and transfers created rapid growth from a low base, but without a later manufacturing-led transformation.

Table 5: Comparator-state real GSDP growth by period

State	1980-1991	1991-2003	2003-2013	2013-latest
Arunachal Pradesh	9.34	5.18	7.45	5.58
Assam	4.17	3.01	4.89	7.71
Sikkim	n.a.	6.54	13.55	7.47
Himachal Pradesh	4.51	6.45	7.87	5.23
Tripura	5.42	7.67	8.33	7.44
Meghalaya	5.87	5.49	7.13	4.34

Notes: Entries are compound annual growth rates in percent. Sikkim has no pre-1991 constant-price GSDP sample in the available data. Latest year is 2024 for all listed states except Sikkim, where it is 2023.

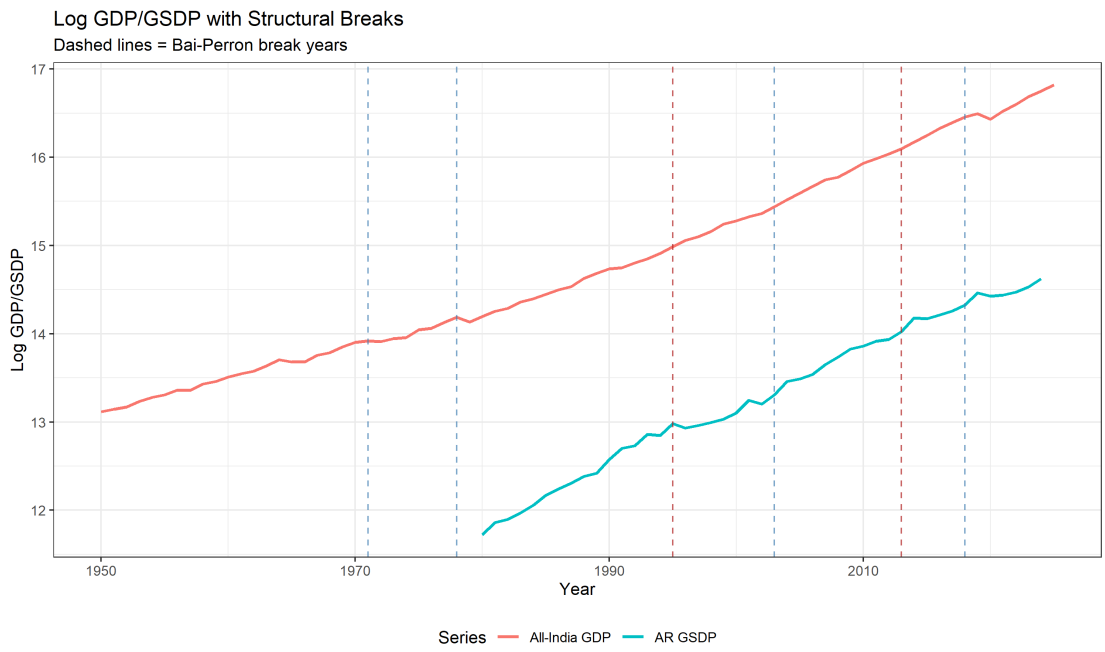

Figure 1: All-India log real GDP with estimated structural breaks

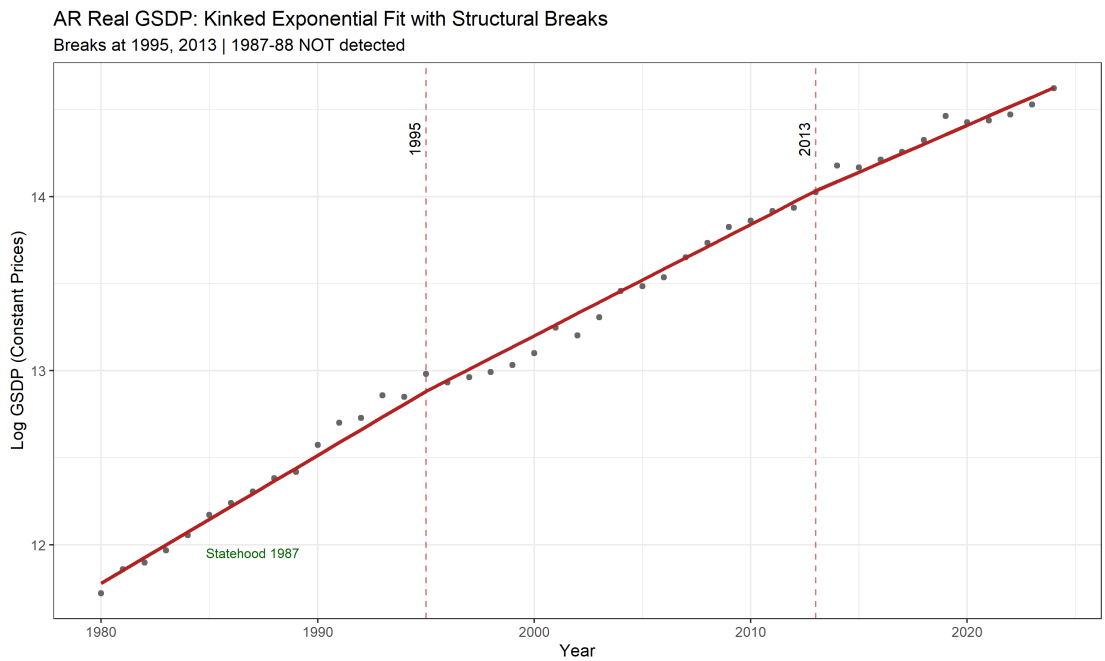

Figure 2: Arunachal Pradesh log real GSDP with estimated structural breaks

6 Structural Transformation

The growth results raise a natural question: if Arunachal Pradesh grew rapidly, what changed inside the economy? The sectoral evidence in Figure 3 points to growth without deep industrial transformation. Agriculture's share declines over the long run, but the space is absorbed mainly by services rather than by a large and durable industrial expansion. By 2024-25, services

account for nearly half of constant-price GSDP, while industry remains modest. This pattern resembles India's broader services-led development path discussed by Kochhar et al. (2006), but in Arunachal Pradesh the relevant service expansion is more plausibly tied to public administration, construction-linked services, transfers, and infrastructure spending than to large-scale private tradable services.

This distinction matters for interpretation. A growth acceleration caused by industrialisation would normally leave signs in the sectoral composition: a rising industrial share, higher labour absorption outside agriculture, and stronger productivity spillovers. Arunachal Pradesh's evidence is weaker on these dimensions. The state grows, but the composition suggests that the growth process remains fiscally and administratively mediated.

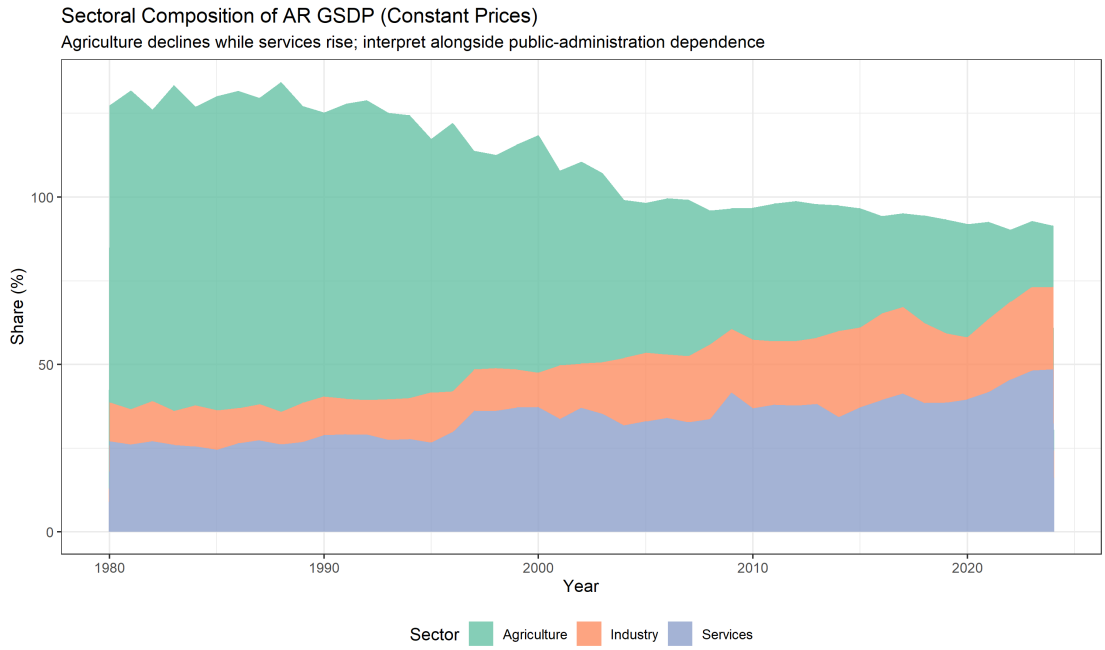

Figure 3: Sectoral shares in Arunachal Pradesh GSDP

7 Inflation Results

Table 6 reports summary statistics for quarterly inflation. Arunachal Pradesh rural CPI inflation averages 5.90 percent, slightly above all-India combined CPI inflation at 5.58 percent. The more important difference is volatility. The standard deviation of Arunachal Pradesh quarterly CPI inflation is 3.34, compared with 2.28 for all India. Deflator-based inflation for Arunachal Pradesh is still more volatile, but that result should be read carefully because quarterly GSDP is interpolated from annual benchmarks.

Table 6: Quarterly inflation summary statistics

Statistic	AR rural CPI	India combined CPI	AR GSDP deflator
Mean	5.90	5.58	6.91
Median	6.01	5.27	5.95
Standard deviation	3.34	2.28	4.34
Minimum	-2.15	0.76	-0.56
Maximum	13.05	10.59	21.24
Number of quarters	56	56	176

The time-series pattern in Figure 4 is consistent with the regional inflation literature. Arunachal Pradesh inflation is not permanently disconnected from the national series, but deviations can be large. The pre-COVID period shows a positive mean premium of 1.22 percentage points. In the COVID and post-COVID period, the premium turns negative on average, partly because all-India inflation remained elevated while Arunachal Pradesh's measured rural headline inflation fell during the pandemic data gap and then normalised.

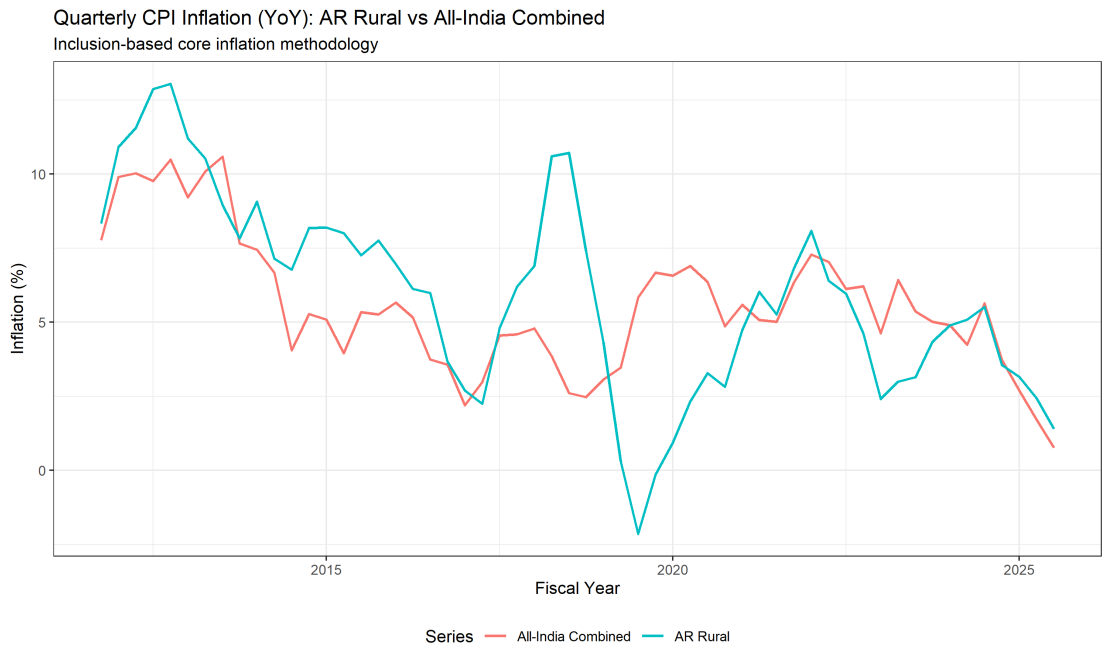

Figure 4: Quarterly CPI inflation: Arunachal Pradesh rural and all-India combined

Annual headline and core inflation are shown in Figure 5 and in full in Appendix Table 9. The annual inflation window begins effectively in 2012-13 because a year-on-year fiscal-year rate for 2011-12 would require a full 2010-11 monthly CPI baseline, which is not available in the source panel. Arunachal Pradesh headline inflation is 12.12 percent in 2012-13 and 8.91 percent in 2018-19, compared with 10.05 and 3.41 percent respectively for all India. In 2020-21, the

Arunachal Pradesh headline series sits well below the all-India rate, but this year must be read with caution because April and May 2020 are missing in the Arunachal Pradesh CPI data. By 2024-25, headline inflation is similar in the two series: 4.76 percent for Arunachal Pradesh and 4.63 percent for all India.

The core comparison needs a more explicit quantitative caveat. Arunachal Pradesh has only a rural CPI and no housing component, whereas housing carries 21.67 urban weight points in the national CPI and 9.83 points, or 22.0 percent, in the population-weighted combined national core basket. Appendix Table 11 therefore reports a sensitivity check in which housing is removed from the national urban core as well. Over 2012-13 to 2024-25, that adjustment changes annual all-India core inflation by 0.29 percentage points on average in absolute value, with a maximum effect of 0.71 points in 2021-22. The sign is not one-sided, because housing inflation sometimes runs below and sometimes above the rest of the core basket. The implication is that the state-national core comparison is imperfect but still informative: the housing omission is large enough to matter for fine year-by-year reading, but not large enough to overturn the paper's broader inflation ranking.

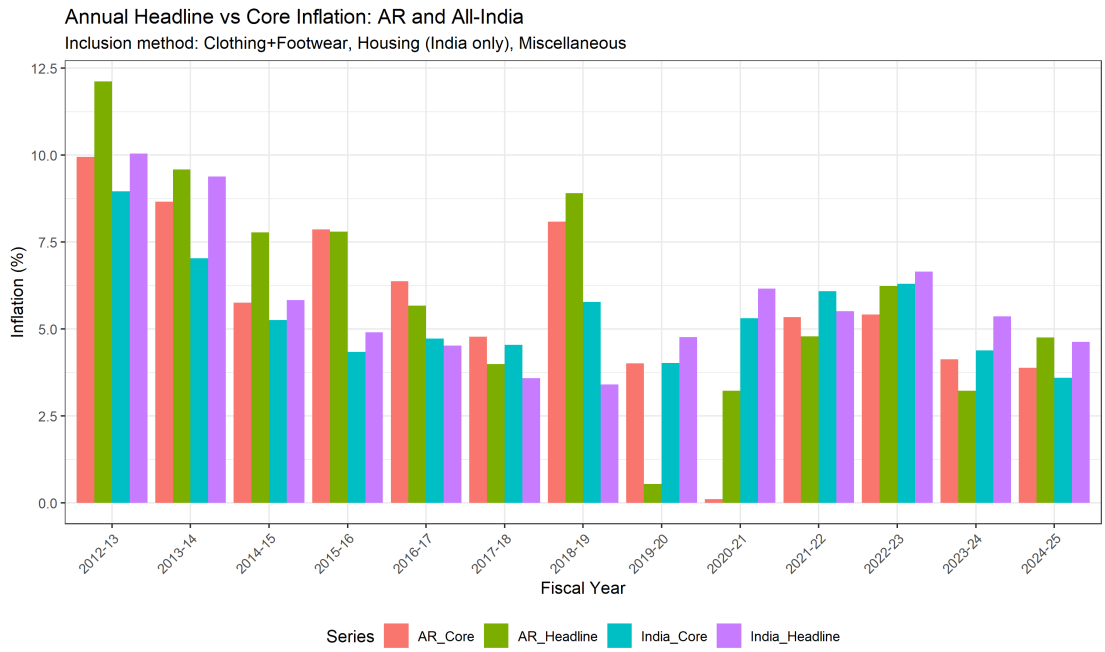

Figure 5: Annual headline and core inflation

The cross-state CPI comparison in Table 7 shows that Arunachal Pradesh is not an outlier in mean headline inflation among Northeast comparators. Its mean headline inflation, 6.05 percent, is close to Tripura and Assam. Its fuel inflation, however, is high: 7.54 percent on average, compared with 5.11 percent for all India and 3.42 percent for Himachal Pradesh. That result deserves separate attention rather than a passing logistics-cost remark.

The degree of national co-movement is also measurable rather than impressionistic. The Pearson correlation between quarterly year-on-year Arunachal Pradesh rural headline inflation and all-India combined headline inflation is 0.54 over the 56-quarter overlap, with $p < 0.001$; Appendix Table 11 reports the formal test. The state is therefore not disconnected from

the national inflation cycle. What distinguishes it is partial rather than zero integration: co-movement is positive, but far from one-for-one, volatility is higher, and the fuel premium remains persistently positive. The same table also shows that the correlation is strong before COVID-19 and much weaker after 2020, which is consistent with pandemic-era supply distortions and the temporary Arunachal Pradesh data gap.

Figure 7 shows that Arunachal Pradesh's annual fuel inflation premium over all India is positive in most years. The premium averages roughly 3.0 percentage points in 2012-13 to 2018-19 and 2.6 points in 2021-22 to 2024-25, so it narrows slightly after the pandemic but does not disappear. The same-year correlation between Arunachal Pradesh fuel inflation and Arunachal Pradesh headline inflation is 0.61, whereas the one-year lead correlation is close to zero. The evidence therefore supports a contemporaneous cost-pass-through reading more strongly than a clean leading-indicator story: when fuel prices spike in the state, headline inflation tends to move with them in the same year. This is exactly what one would expect in a thin, high-transport-cost market where fuel is an input into retail distribution as much as a household purchase item. The pandemic year should again be read cautiously because missing months depress the state series mechanically.

Table 7: Cross-state CPI pressure, 2012-13 to 2024-25

State	Headline mean	Headline SD	Food mean	Fuel mean
All India	5.75	1.97	6.16	5.11
Arunachal Pradesh	6.05	3.16	5.87	7.54
Assam	5.87	2.14	5.75	8.12
Himachal Pradesh	5.20	2.36	5.47	3.42
Meghalaya	5.34	3.68	5.34	7.11
Sikkim	5.83	2.20	6.07	7.98
Tripura	6.06	3.33	5.79	8.84

Notes: Entries are annual inflation rates in percent. Arunachal Pradesh uses rural CPI; other states use combined CPI where available. Headline, food, and fuel inflation are read directly from the published combined CPI group indices. Exact combined-state core CPI is not reported because the state weight file supplies rural and urban basket weights, but not a state-specific combined basket weight vector.

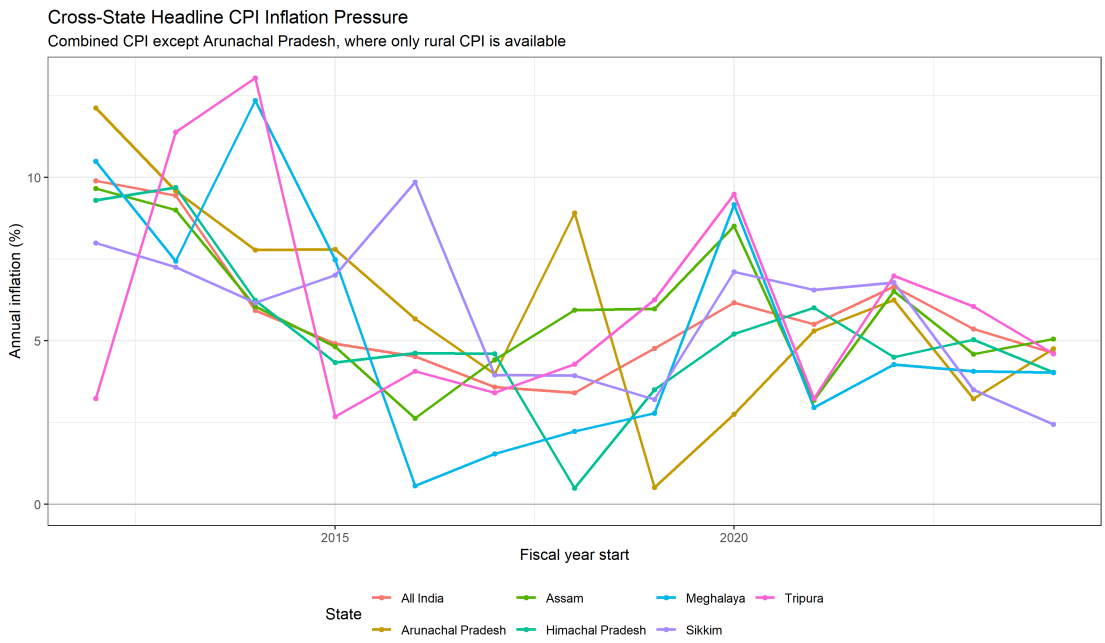

Figure 6: Cross-state CPI pressure

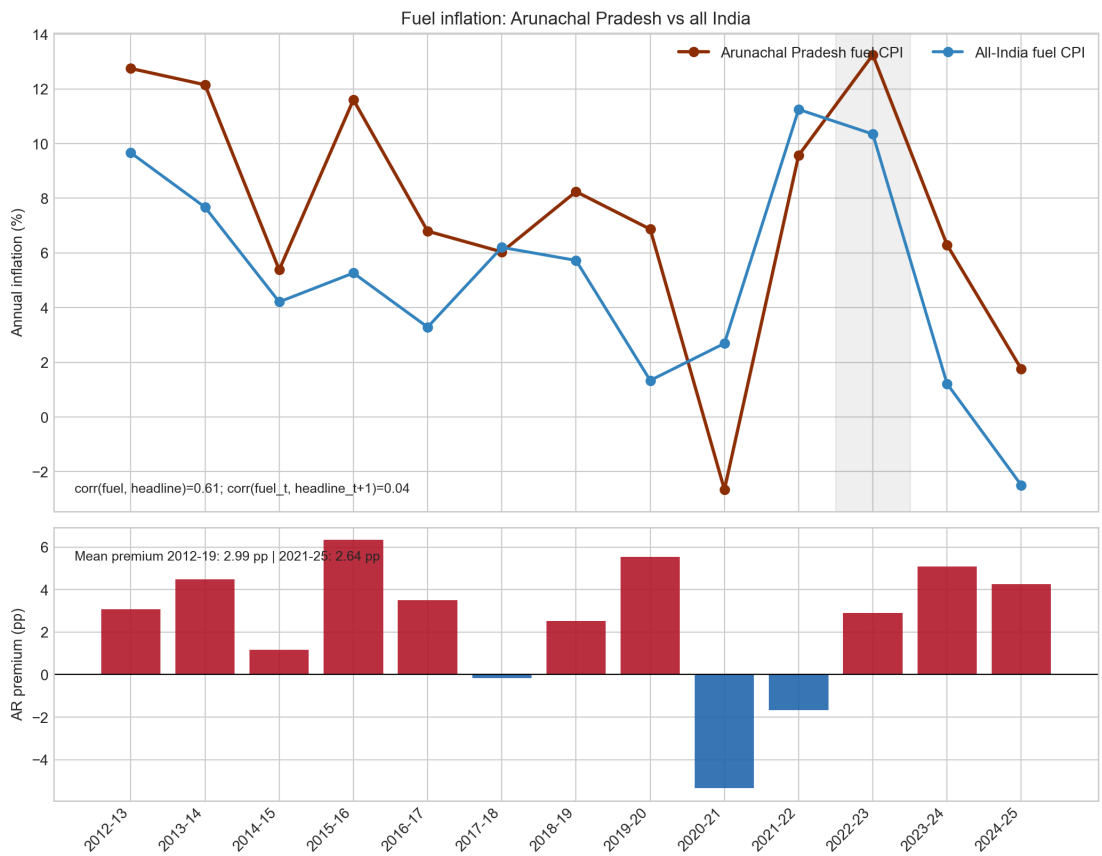

Figure 7: Arunachal Pradesh fuel inflation and the premium over all India

8 COVID-19 Shock and Recovery

The formal COVID-19 pulse dummy for Arunachal Pradesh is negative but not statistically significant at conventional levels: the coefficient is -0.0344 with a robust standard error of 0.0209 and a p-value of 0.107. This does not imply that there was no shock. It means that, relative to the fitted trend and the short annual sample, the one-year level shift is imprecisely estimated.

The realised-shock comparison in Table 8 is more transparent. Arunachal Pradesh's real GSDP fell by 3.69 percent from 2019-20 to 2020-21, compared with a pre-COVID trend CAGR of 5.88 percent. The shock relative to trend is therefore -9.57 percentage points. The fall is milder than Meghalaya's -7.85 percent actual contraction and similar in broad magnitude to Himachal Pradesh and Tripura. The recovery, however, is comparatively weak. Arunachal Pradesh's post-2020 recovery CAGR is 5.02 percent, below Assam, Sikkim, Tripura, and Meghalaya.

That weak rebound fits the composition of the state economy. Recent macro-fiscal evidence shows a structure dominated by services and agriculture, with manufacturing accounting for only 2.2 percent of workers (NITI Aayog and NCAER 2025). That is not the profile of a state with a large private tradables base ready to snap back after lockdowns. **?** show that COVID-era state heterogeneity in India was partly associated with manufacturing intensity and migration, and they document that Arunachal Pradesh was among the states with an April 2020 electricity decline above 20 percent. In Arunachal Pradesh, transfers and public expenditure can cushion the downturn, but they do not generate the same rebound arithmetic as a broader private-sector base. That helps explain why the initial contraction was not the worst in the sample, yet the subsequent recovery remained slower than in Assam, Tripura, Sikkim, and Meghalaya.

Table 8: COVID-19 growth shock and recovery

State	Pre-COVID trend	Actual 2019-20 to 2020-21	Shock vs trend	Recovery CAGR
Arunachal Pradesh	5.88	-3.69	-9.57	5.02
Assam	7.82	2.95	-4.86	9.00
Sikkim	8.32	0.33	-7.99	8.39
Himachal Pradesh	6.36	-4.40	-10.76	5.80
Tripura	7.14	-4.36	-11.50	8.40
Meghalaya	4.35	-7.85	-12.20	9.54

Notes: Entries are percent. Recovery CAGR is from 2020-21 to the latest available year, 2024-25 except Sikkim, where the latest available year is 2023-24.

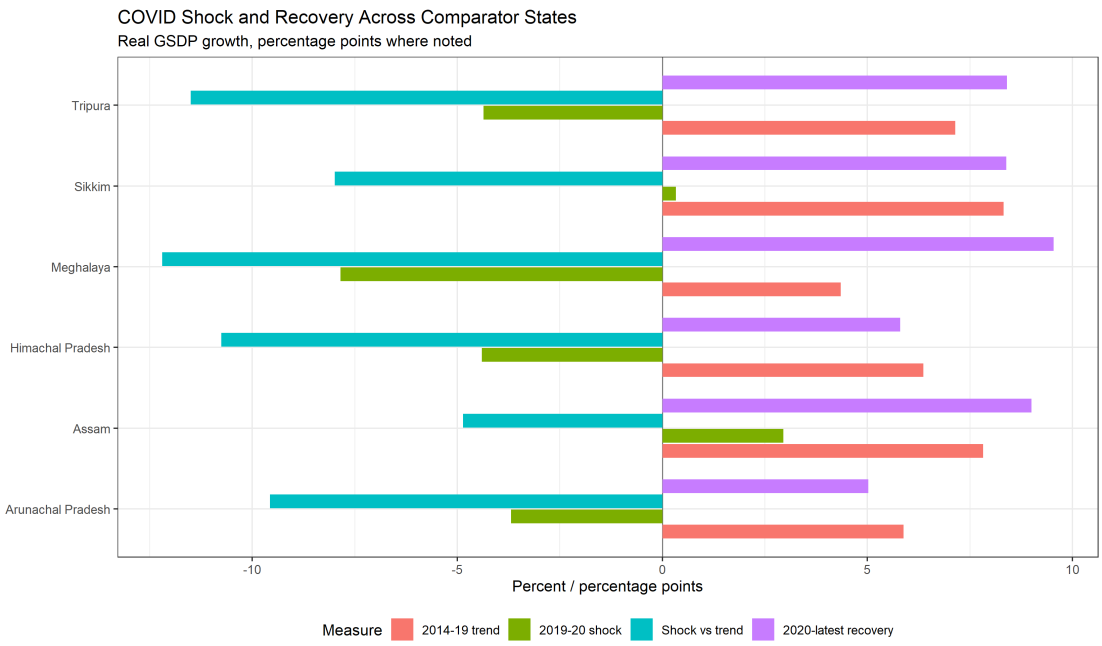

Figure 8: COVID-19 shock and recovery across comparator states

9 Robustness and Comparator Evidence

Three robustness exercises support the main interpretation. First, the Arunachal Pradesh break dates are stable to alternative trimming values. With trimming from 0.10 to 0.20, the state continues to show two breaks, dated 1995 and 2013. All-India breaks are more sensitive: the baseline four-break result simplifies to 1978 and 2004 at higher trimming values. This strengthens the caution that India's aggregate growth transition should not be reduced to a single date.

Second, the alternative growth specification does not overturn the qualitative result, but it clarifies the econometric stakes. Appendix Figures 12 and 13 overlay the continuity-restricted and discontinuous fits. For Arunachal Pradesh, the two specifications trace similar broad trajectories, but the segment-wise fit sits above the continuity-restricted line around the 1995-96 break and below it in the final regime. This is consistent with Table 4: the main disagreement is not whether growth slowed, but whether the 1995 break should be read purely as a kink in trend growth or partly as a level re-alignment.

Third, the state-comparator module shows that Arunachal Pradesh is a high-growth state from a low base, but not the strongest post-2003 or post-COVID performer. Figure 9 indexes real GSDP paths, while Figure 10 compares liberalisation-era growth. Sikkim is the most dramatic small-state high-growth benchmark after 2003. Assam is weaker before 2013 but outperforms Arunachal Pradesh after 2013. Tripura's 1991-2003 and 2003-13 performance is stronger than Arunachal Pradesh's. Appendix Figure 14 is consistent with the broader divergence literature: cross-state dispersion in log per-capita NSDP does not show a clean monotonic decline, which is exactly why unconditional catch-up is the wrong prior for a frontier state (Sachs et al. 2002; Bhattacharya and Sakthivel 2004; **?**). These comparisons sharpen the paper's thesis: Arunachal Pradesh's growth is impressive in levels and early growth rates, but it is not a uniquely successful

post-reform growth acceleration.

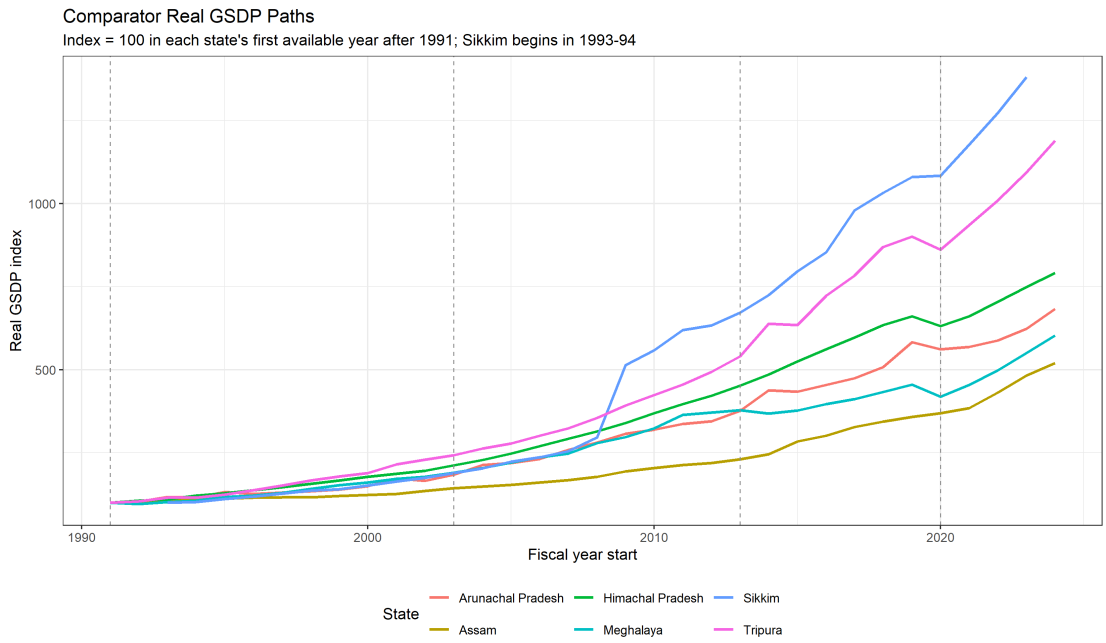

Figure 9: Indexed real GSDP paths across comparator states

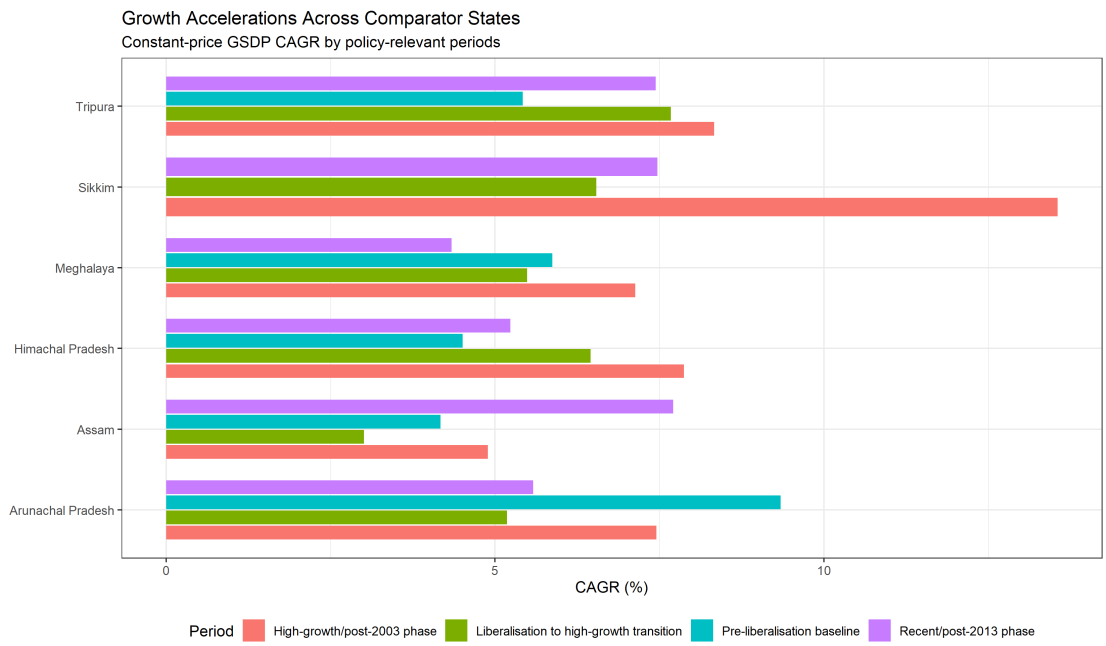

Figure 10: Growth comparison around liberalisation and later regimes

10 Discussion

The results suggest three substantive conclusions. First, Arunachal Pradesh's growth history is not well described by the standard all-India liberalisation narrative. The state does not show a 1991 break, and its pre-1991 growth rate is higher than its 1991-2003 growth rate. This is

consistent with the literature arguing that India's growth transition itself is more complex than a single reform date, and it is even more relevant for a frontier state whose economic base differs sharply from the national average.

Second, growth has not produced deep structural transformation. The sectoral shift away from agriculture is real, but the receiving sector is mainly services. In Arunachal Pradesh this likely reflects public administration, construction-linked activity, and transfer-financed services more than autonomous private-sector dynamism. The fiscal evidence should therefore be read as corroboration, not as decoration. Appendix Figure 15 shows a fiscal dependence ratio averaging 87.4 percent, while Appendix Figure 18 shows that own revenue remains modest relative to revenue expenditure. Combined with the services-heavy output structure and the very small manufacturing employment share, those fiscal patterns make the transfer-dependent label descriptive rather than rhetorical (Govil and Chauhan 2024; NITI Aayog and NCAER 2025). The main causal evidence in this paper still comes from break timing, sectoral composition, and inflation behavior rather than from fiscal ratios alone.

Third, inflation behaves like a partially integrated frontier-market phenomenon. Arunachal Pradesh does not move independently of all India: quarterly headline CPI inflation has a positive correlation of 0.54 with the national series. But the relationship is far from one-for-one, volatility is higher, and fuel inflation is especially elevated. This is exactly where terrain and logistics should matter. The CPI evidence also warns against over-precision. The absence of urban CPI, the missing 2020 months, and the narrow state core-inflation measure all limit comparability. Even so, the broad result is robust: Arunachal Pradesh faces more volatile consumer-price pressure than all India, and its inflation experience must be read through food, fuel, and transport-cost channels.

11 Conclusion

This paper provides a complete empirical assessment of Arunachal Pradesh's growth and inflation performance under the assignment brief. The evidence shows a state that grew rapidly but did not experience a standard post-1991 acceleration. Its structural breaks are dated at 1995-96 and 2013-14; its trend growth declines across regimes; and its pre-liberalisation growth exceeds its 1991-2003 growth. A discontinuous segment-wise sensitivity check does not overturn that broad deceleration result, but it does suggest that the 1995-96 break may combine a level re-alignment with a slope change. Inflation is slightly higher on average and clearly more volatile than all India, with a marked fuel component whose premium persists even after the pandemic. The COVID-19 output shock was real but not the most severe among comparable hill and Northeast states, while the recovery has been weaker than several comparators.

The best interpretation is therefore not simply that Arunachal Pradesh is a high-growth state. It is that Arunachal Pradesh is a transfer-dependent frontier economy: public expenditure and central transfers sustain demand and services-led output expansion, while terrain, thin markets, and limited industrial transformation shape inflation and long-run growth constraints. The policy implication is that macroeconomic performance should be judged not only by the level of

GSDP growth, but also by the source, composition, volatility, and resilience of that growth.

A Additional Tables

Table 9: Annual headline and core inflation

Fiscal year	AR headline	AR core	India headline	India core
2011-12	n.a.	n.a.	n.a.	n.a.
2012-13	12.12	9.95	10.05	8.96
2013-14	9.59	8.66	9.38	7.03
2014-15	7.78	5.76	5.83	5.26
2015-16	7.80	7.86	4.91	4.34
2016-17	5.67	6.37	4.52	4.72
2017-18	3.99	4.78	3.59	4.54
2018-19	8.91	8.09	3.41	5.78
2019-20	0.54	4.01	4.77	4.02
2020-21	3.23	0.11	6.16	5.31
2021-22	4.79	5.34	5.51	6.09
2022-23	6.24	5.42	6.65	6.30
2023-24	3.22	4.13	5.36	4.38
2024-25	4.76	3.88	4.63	3.60

Table 10: Inflation before and after COVID-19

Period	AR mean	AR SD	India mean	India SD	N
Pre-COVID, 2012-2019	7.10	3.65	5.87	2.61	33
COVID and post-COVID, 2020 onward	4.18	1.82	5.15	1.66	23

Table 11: Inflation comparability and co-movement diagnostics

Diagnostic	Value	Note
Housing weight in the population-weighted combined national core basket	22.01	percent of combined core basket
Housing weight in the population-weighted combined national core basket	9.83	weight points
Mean absolute change in annual all-India core inflation when housing is excluded, 2012-13 to 2024-25	0.29	percentage points
Maximum absolute change in annual all-India core inflation when housing is excluded	0.71	percentage points, in 2021-22
Pearson correlation: quarterly Arunachal Pradesh vs all-India headline inflation, full overlap	0.54	$p < 0.001$, $N = 56$
Pearson correlation: quarterly Arunachal Pradesh vs all-India headline inflation, pre-COVID	0.70	$p < 0.001$, $N = 29$
Pearson correlation: quarterly Arunachal Pradesh vs all-India headline inflation, 2020 onward	0.24	$p = 0.221$, $N = 27$

Table 12: Break-date trimming sensitivity

Trimming	India breaks	India break years	AR breaks	AR break years
0.10	4	1971, 1978, 2003, 2018	2	1995, 2013
0.12	2	1978, 2004	2	1995, 2013
0.15	2	1978, 2004	2	1995, 2013
0.18	2	1978, 2004	2	1995, 2013
0.20	2	1978, 2004	2	1995, 2013

Table 13: Comparator data coverage audit

State	Role	GSDP first	GSDP last	GSDP miss.	PC first	PC last	CPI miss.	CPI sector and caveat
Arunachal Pradesh	Assigned frontier state	1980	2024	0	1970	2024	2	Rural; exact cross-state core not computed
Assam	Northeast regional baseline	1980	2024	0	1970	2024	2	Combined; exact cross-state core not computed
Himachal Pradesh	Hill-state counterfactual	1980	2024	0	1967	2024	2	Combined; exact cross-state core not computed
Manipur	Data-audit NE comparator	1980	2023	1	1960	2023	3	Combined; exact cross-state core not computed
Meghalaya	Northeast robustness comparator	1980	2024	0	1980	2024	2	Combined; exact cross-state core not computed
Mizoram	Data-audit NE comparator	1999	2023	9	1999	2023	2	Combined; exact cross-state core not computed
Nagaland	Data-audit NE comparator	1993	2023	3	1980	2023	3	Combined; exact cross-state core not computed
Sikkim	Small Himalayan benchmark	1993	2023	3	1993	2023	2	Combined; exact cross-state core not computed
Tripura	Northeast robustness comparator	1980	2024	0	1960	2024	2	Combined; exact cross-state core not computed
Uttarakhand	Hill-state robustness comparator	1980	2024	0	1993	2024	2	Combined; exact cross-state core not computed

Table 14: Recent Arunachal Pradesh fiscal indicators

Fiscal year	Figure type	Rev. bal./GSDP	Fiscal bal./GSDP	Primary bal./GSDP	Interest/Rev. exp.
2021-22	Account	16.47	3.25	5.63	4.91
2022-23	Account	17.84	-0.79	1.55	4.79
2023-24	Account	17.83	8.06	10.29	4.17
2024-25	Revised	16.30	2.90	5.04	3.60
2025-26	Budget	9.44	2.61	4.66	3.32

Notes: This table is included as supporting fiscal context for the transfer-dependent growth interpretation. Positive fiscal balance denotes surplus.

B Additional Figures

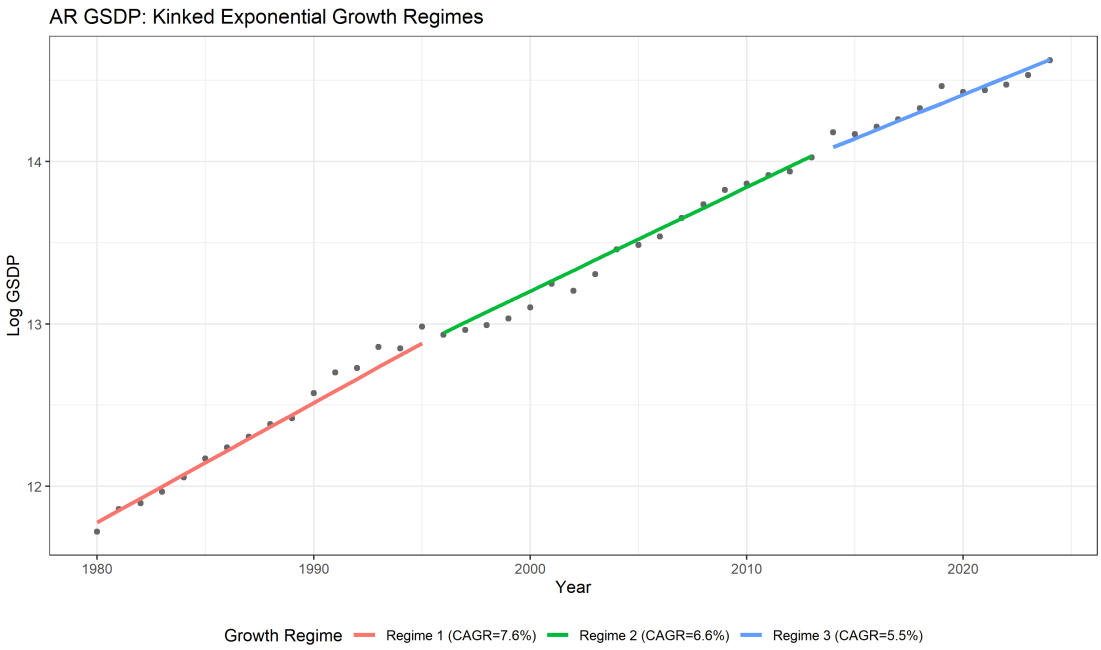

Figure 11: Kinked exponential fit for Arunachal Pradesh

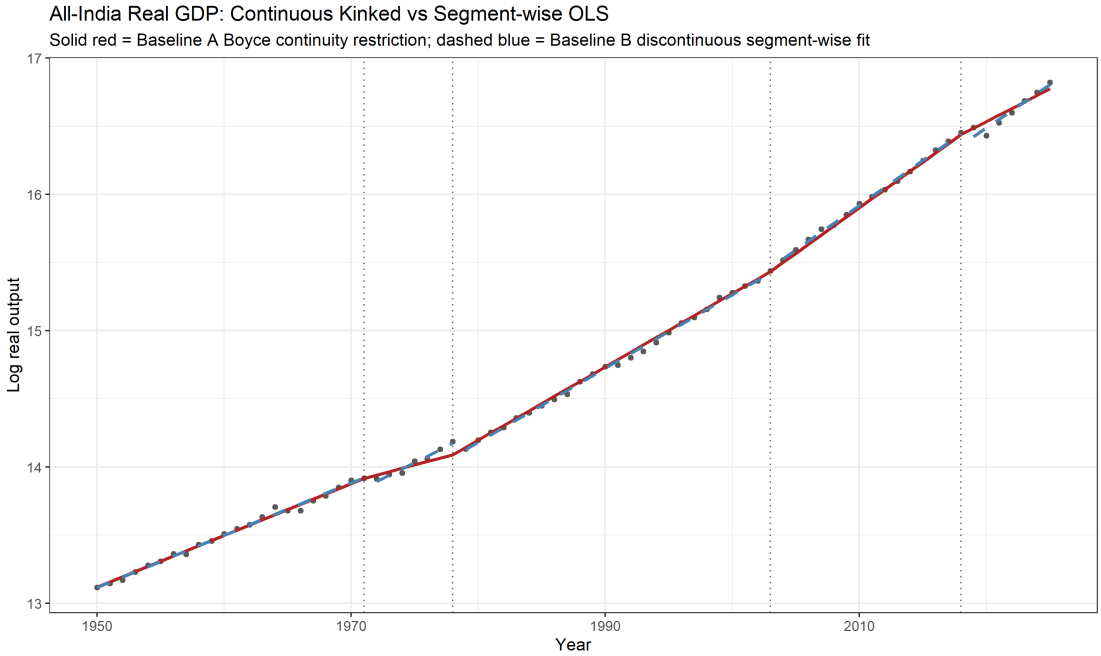

Figure 12: All-India continuous kinked fit and segment-wise OLS fit

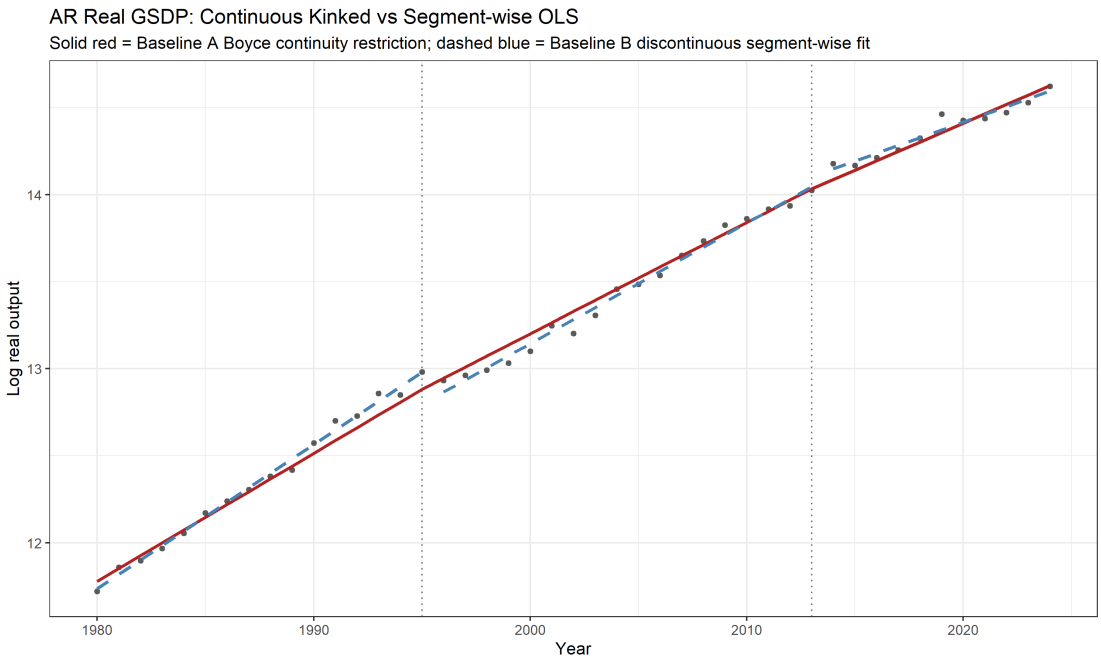

Figure 13: Arunachal Pradesh continuous kinked fit and segment-wise OLS fit

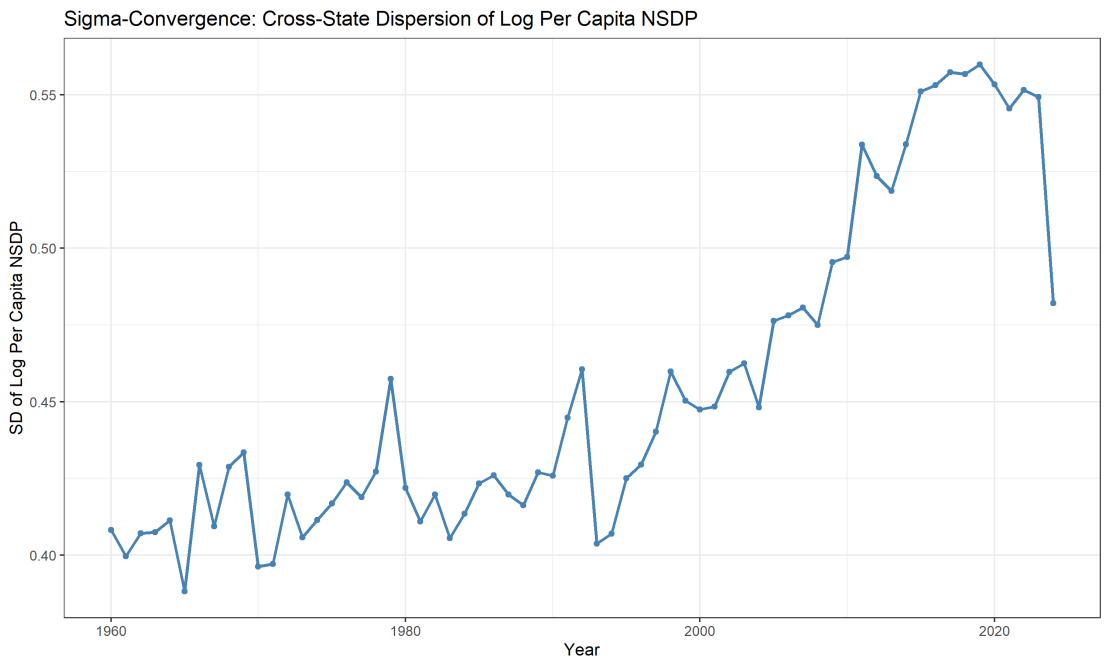

Figure 14: Sigma convergence across Indian states

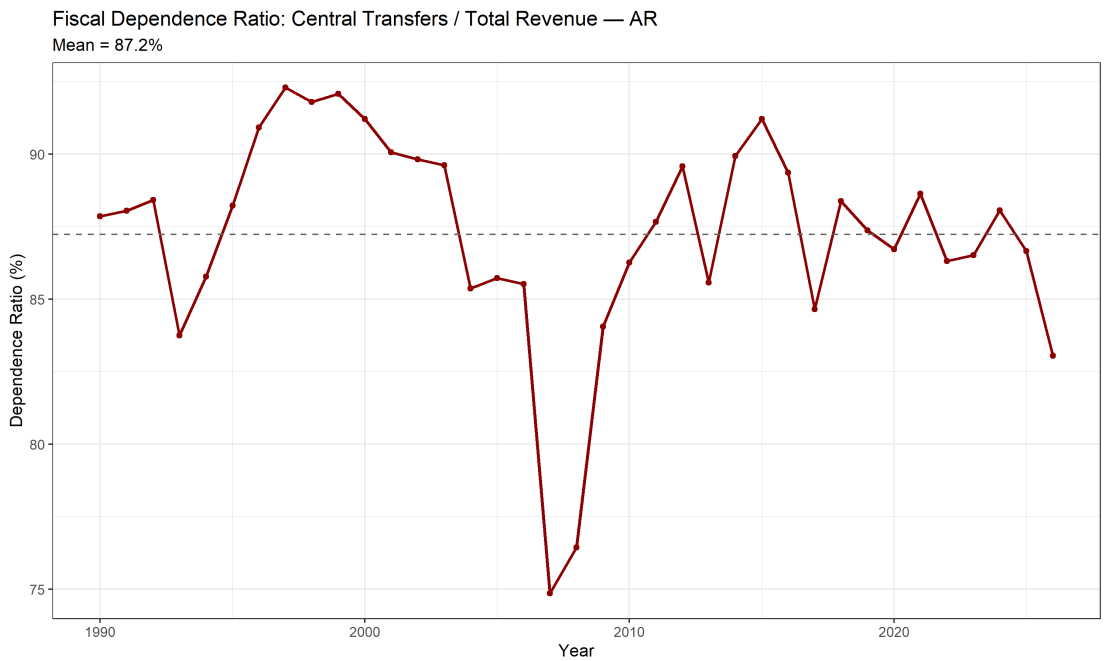

Figure 15: Fiscal dependence ratio: related context

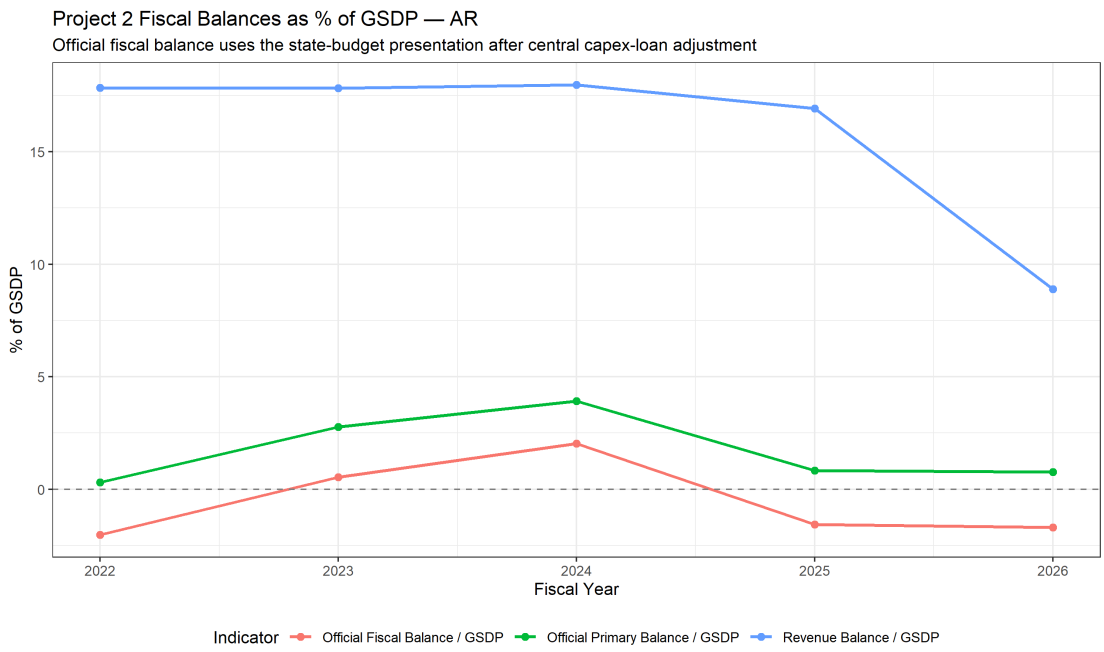

Figure 16: Fiscal balances as share of GSDP: related context

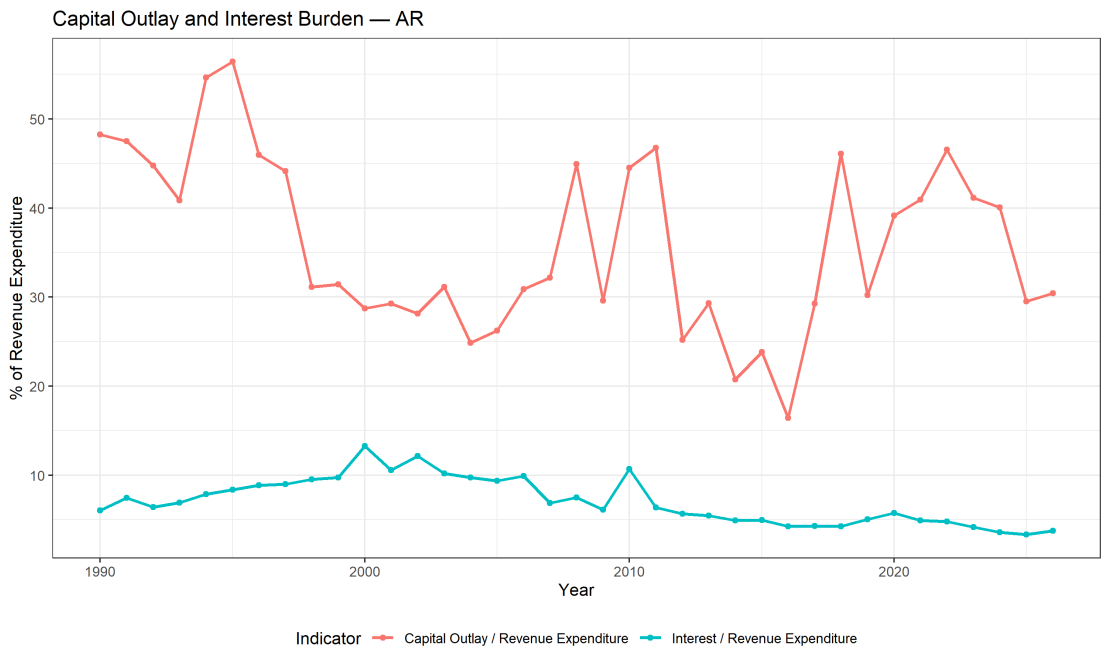

Figure 17: Interest payments as share of revenue expenditure: related context

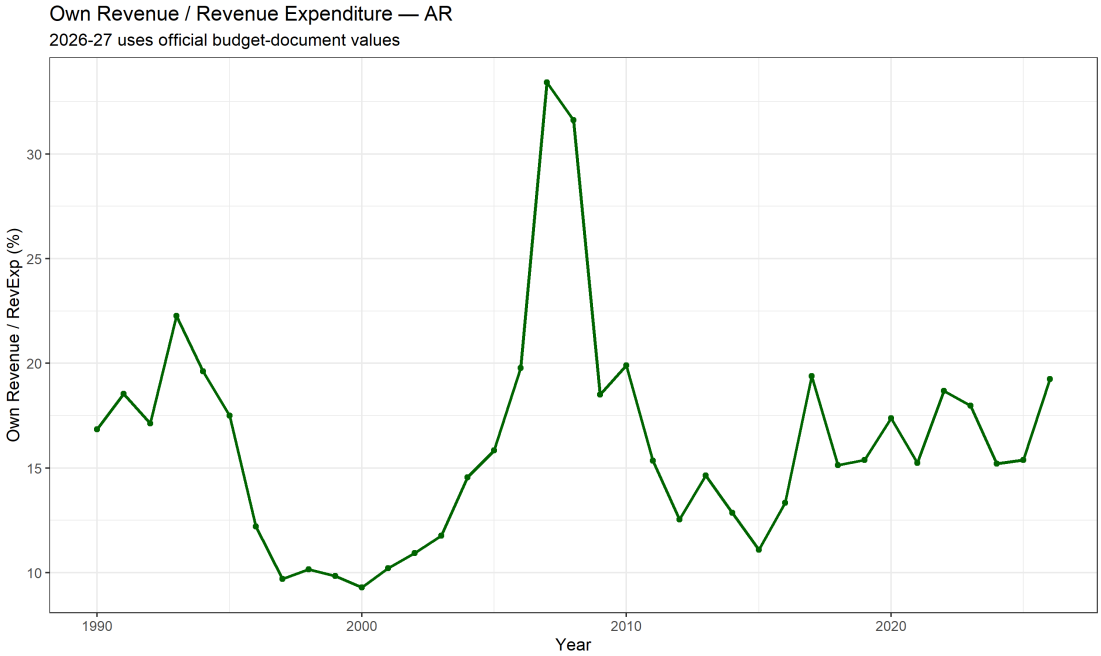

Figure 18: Own revenue as share of revenue expenditure: related context

C Data Limitations

Several limitations shape interpretation. First, Arunachal Pradesh GSDP before February 1987 refers to the territory that became the state. Second, Arunachal Pradesh CPI is rural-only. Urban CPI is absent and combined CPI is not reliable, so the state inflation analysis cannot be a rural-urban combined index like all India. Third, the state CPI basket is available only at group level, and housing is absent from the rural CPI series; the core index is therefore narrower than all-India core inflation. Fourth, April and May 2020 CPI observations are missing for Arunachal Pradesh, affecting 2020-21 annual and quarterly calculations. Fifth, quarterly GSDP deflator inflation is derived using Denton-Cholette interpolation from annual data, so the annual trend is reliable but sub-annual deflator movement should not be over-interpreted. Sixth, annual fiscal-year inflation effectively starts in 2012-13 because the monthly CPI panel begins only in January 2011, leaving no full 2010-11 fiscal-year baseline for a 2011-12 year-on-year calculation. Seventh, cross-state headline, food, and fuel inflation can be computed from the published combined CPI indices, but an exact combined-state core inflation measure is not reported because the available state weight file provides rural and urban basket weights rather than a state-specific combined basket weight vector.

References

Agarwal, M. and Whalley, J. (2013). The 1991 reforms, indian economic growth, and social progress. Technical Report 19024, National Bureau of Economic Research.

Bai, J. and Perron, P. (1998). Estimating and testing linear models with multiple structural changes. *Econometrica*, 66(1):47–78.

Bai, J. and Perron, P. (2003). Computation and analysis of multiple structural change models. *Journal of Applied Econometrics*, 18(1):1–22.

Balakrishnan, P. and Parameswaran, M. (2007). Understanding economic growth in india: A prerequisite. *Economic and Political Weekly*, 42(27):2915–2922.

Bandyopadhyay, S. (2011). Rich states, poor states: Convergence and polarisation in india. *Scottish Journal of Political Economy*, 58(3):414–436.

Beyer, R. C. M., Franco-Bedoya, S., and Galdo, V. (2021). Examining the economic impact of covid-19 in india through daily electricity consumption and nighttime light intensity. *World Development*, 140:105287.

Bhattacharya, B. B. and Sakthivel, S. (2004). Regional growth and disparity in india: A comparison of pre- and post-reform decades. *Economic and Political Weekly*, 39(10):1071–1077.

Boyce, J. K. (1986). Kinked exponential models for growth rate estimation. *Oxford Bulletin of Economics and Statistics*, 48(4):385–391.

Cashin, P. and Sahay, R. (1995). Internal migration, center-state grants and economic growth in the states of india. Technical Report WP/95/66, International Monetary Fund.

Fourteenth Finance Commission (2015). Report of the fourteenth finance commission. Technical report, Government of India.

Govil, R. and Chauhan, K. (2024). Growth convergence and public finances of india and its states. Technical Report WP/24/235, International Monetary Fund.

Hausmann, R., Pritchett, L., and Rodrik, D. (2004). Growth accelerations. Technical Report 10566, National Bureau of Economic Research.

Jha, A. and Dhal, S. C. (2019). Spatial inflation dynamics in india: An empirical perspective. *Reserve Bank of India Occasional Papers*, 40(1).

Kochhar, K., Kumar, U., Rajan, R., Subramanian, A., and Tokatlidis, I. (2006). India's pattern of development: What happened, what follows? Technical Report WP/06/22, International Monetary Fund.

Kundu, S., Kishore, V., and Bhoi, B. B. (2018). Regional inflation dynamics in india. *RBI Bulletin*.

Mahajan, K. and Tomar, S. (2021). Covid-19 and supply chain disruption: Evidence from food markets in india. *American Journal of Agricultural Economics*, 103(1):35–52.

Ministry of Home Affairs, Government of India (1996). Annual report 1995–96. North Eastern Council section documents road, power, and regional-institution support in the Northeast.

Narayanan, S. and Saha, S. (2021). India's food supply chain during the pandemic. *Food Policy*, 105:102162.

NITI Aayog and NCAER (2025). Macro and fiscal brief: Arunachal pradesh. March 2025 state macro-fiscal brief.

Rodrik, D. and Subramanian, A. (2005). From hindu growth to productivity surge: The mystery of the indian growth transition. *IMF Staff Papers*, 52(2):193–228.

Sachs, J. D., Bajpai, N., and Ramiah, A. (2002). Understanding regional economic growth in india. *Asian Economic Papers*, 1(3):32–62.

Sodsriwiboon, P. and Kalra, S. (2010). Growth convergence and spillovers among indian states: What matters? what does not? Technical Report WP/10/96, International Monetary Fund.

Zeileis, A., Leisch, F., Hornik, K., and Kleiber, C. (2002). strucchange: An r package for testing for structural change in linear regression models. *Journal of Statistical Software*, 7(2):1–38.